

Vidhooor Legal AI Copilot - Technical Documentation

Executive Summary

Vidhooor Legal AI Copilot is an intelligent legal research platform designed to bridge the gap between complex Indian legal frameworks and common citizens, while empowering law students and firms with an ultra-high-speed research copilot.

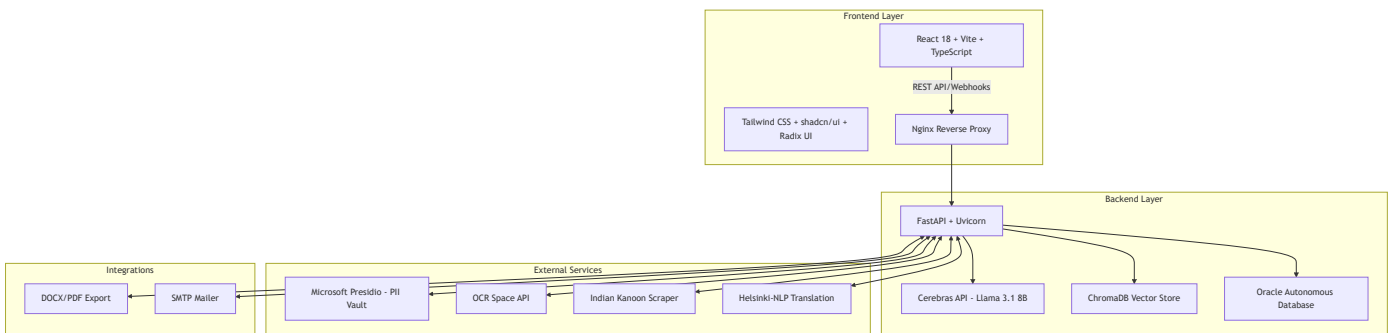
Impact Metrics

- **Inference Speed:** 2,350 tokens/second on Cerebras hardware
- **Legal Corpus:** 851 pages of Indian legal frameworks indexed
- **Time Reduction:** 4-6 hour research tasks reduced to ~30 seconds
- **Efficiency Gain:** Eliminates ~30% of manual cross-referencing time for legal professionals

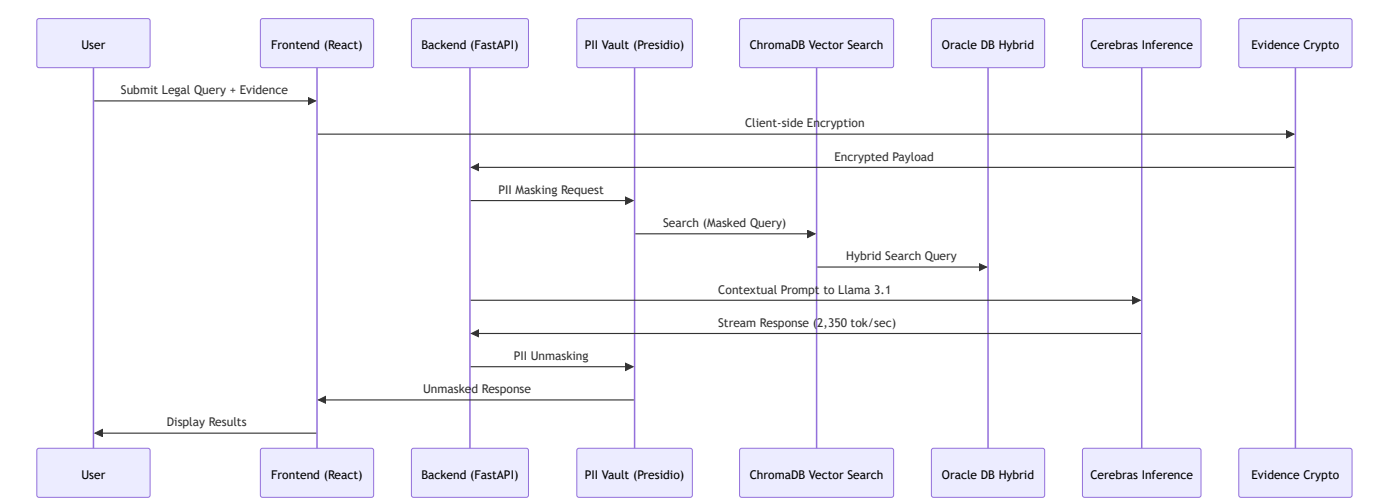
Tech Stack: FastAPI, Llama 3.1, ChromaDB, Oracle Autonomous Database, React 18, Vite, TypeScript, Tailwind CSS, Nginx, Docker, OCI

System Architecture

Core Architecture Flow



RAG & Security Sequential Flow



Feature Comparison Matrix

Feature/Aspect	Generic AI (ChatGPT)	Legacy Legal DBs	Vidhoor Legal AI Copilot
Legal Context/Hallucinations	High risk of hallucinations, no domain-specific training	Exact text matching only, no semantic understanding	Low hallucination risk, trained on 851 pages of Indian law
Inference Latency	~100-200 tokens/sec typical	Instant search but no generation	2,350 tokens/sec via Cerebras
Data Privacy/PII Protection	Standard cloud privacy, no PII-specific handling	Basic database security	Enterprise PII masking (Presidio) + client-side encryption
Accessibility/Language Complexity	English-focused	Complex legal jargon, English only	Multi-lingual support via Helsinki-NLP translation

Feature/Aspect	Generic AI (ChatGPT)	Legacy Legal DBs	Vidhoor Legal AI Copilot
Workflow Automation	Manual copy-paste workflows	Manual document compilation	Automated drafting to PDF/DOCX + SMTP mail delivery

Code Module Breakdown

Backend Structure

Module	Purpose	Key Components
main.py	FastAPI application entry point, route definitions	API endpoints, middleware setup
chroma_manager.py	Vector database management	Embedding storage, metadata-aware citation tracking, similarity search
database.py	Oracle Autonomous Database connector	User sessions, chat history, encrypted evidence storage
pii_vault.py	PII masking/unmasking service	Microsoft Presidio integration, automated redaction
services/	Business logic services	OCR processing, web scraping, translation pipelines
ingest_*.py	Data ingestion pipelines	Legal document parsing, embedding generation
ocr_vision.py	Optical Character Recognition	FIR scanning, document text extraction via OCR Space API

Module	Purpose	Key Components
<code>indian_kanoon_live.py</code>	Live case-law scraping	Real-time legal precedent mining
<code>translate_helsinki.py</code>	Multi-lingual support	Hindi/English translation via Hugging Face models
<code>draft_exporter.py</code>	Legal document generation	PDF/DOCX export compilation
<code>draft_mailer.py</code>	Automated delivery	SMTP integration for draft sharing

Frontend Structure

Component	Purpose	Key Features
<code>ChatArea.tsx</code>	Main conversation display	React Markdown rendering, citation display
<code>ChatInput.tsx</code>	User interaction input	Evidence upload, query submission
<code>evidenceCrypto.ts</code>	Client-side encryption	E2EE before transmission to backend
<code>components/ui/*</code>	UI primitives	shadcn/ui + Radix UI component library
<code>App.tsx</code>	Application shell	Routing, query client, tooltip provider

Deployment & Security Architecture

Security Mechanisms

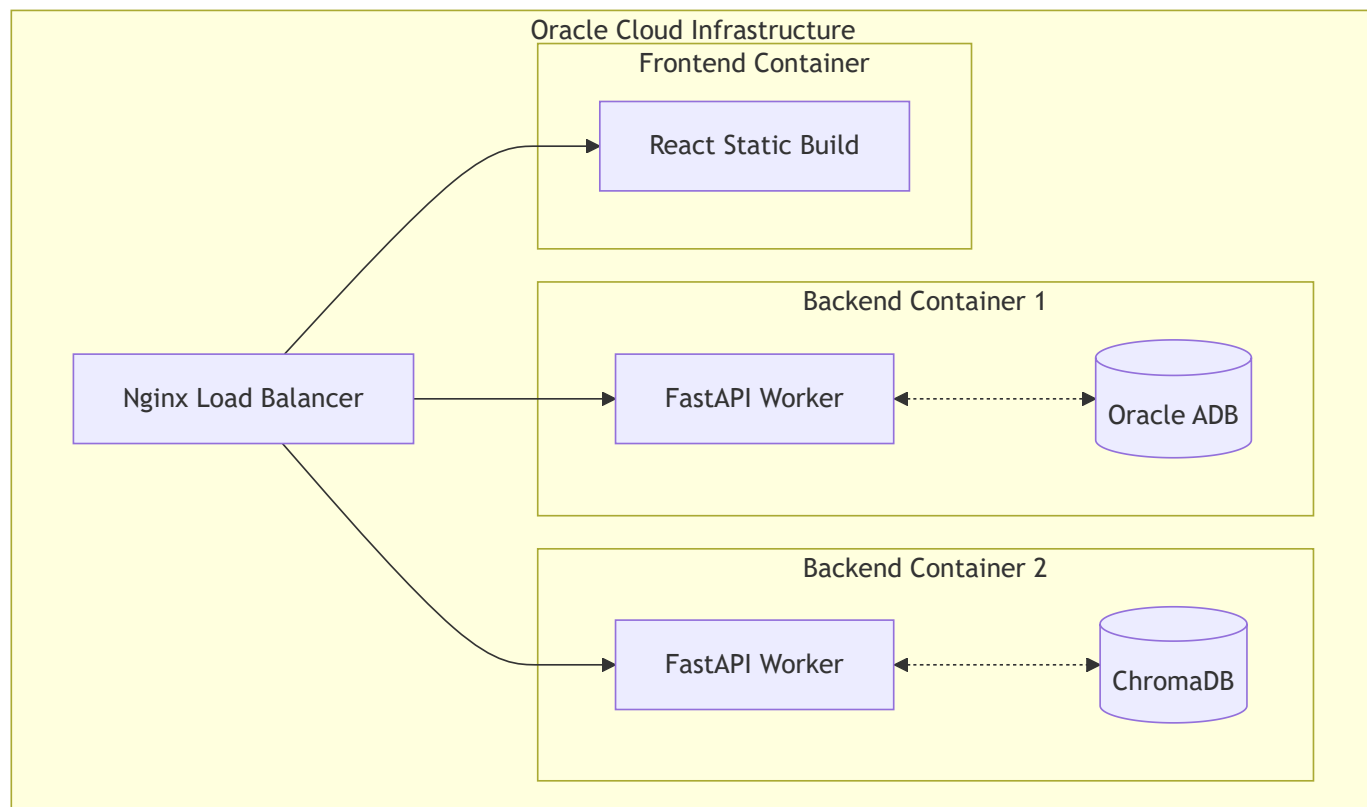
1. PII Protection Pipeline

- Microsoft Presidio automatic entity detection and masking
- PII vault for secure storage of sensitive legal data
- Client-side encryption via `evidenceCrypto.ts` before transmission

2. End-to-End Encryption

- Evidence files encrypted in browser before upload
- Encrypted payloads stored in Oracle Autonomous Database
- Decryption only happens after PII-safe processing

Production Runtime Topology



Containerized Deployment

- **Docker Compose:** `docker-compose.oci.yml` orchestrates multi-container setup
- **Nginx:** Reverse proxy with SSL termination, static file caching
- **OCI Deployment Scripts:**
 - `build_and_push_oci.sh`: Container build and registry push
 - `deploy_backend_oci.sh`: Backend deployment orchestration

Business & Monetization Model

Revenue Distribution Matrix

Tier	Target Audience	Revenue Weight	Features
B2B SaaS	Law Firms, Legal Professionals	50%	Full feature access, priority support, custom integrations
Academic Tier	Law Students, Researchers	30%	Discounted access, educational resources, research tools
B2C Freemium	General Public	20%	Basic queries, limited evidence analysis, public legal literacy

Key Technical Highlights

- **Ultra-low Latency:** 2,350 tokens/second inference on Cerebras hardware API
- **Localized Legal Corpus:** 851 pages of Indian legal frameworks (Constitution, BNS, BNSS, BSA, IPC, IT Act, DPDP Act)
- **Hybrid Database Strategy:** ChromaDB for vector search + Oracle ADB for persistent state
- **Real-time Integration:** Live scraping of Indian Kanoon with contextual relevance scoring
- **Document Intelligence:** OCR pipeline for FIR and legal paper ingestion
- **Multi-lingual Support:** Helsinki-NLP translation pipelines for Hindi/English accessibility